

A Study of Cross-Session Cross-Device Search Within an Academic Digital Library

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ABSTRACT

Information seeking in an academic digital library is complex in nature, often spanning multiple search sessions. Resuming academic search tasks requires significant cognitive effort as searchers must reacquaint themselves with previous search session activities and previously discovered documents before resuming their search. Further, some academic searchers may find it convenient to initiate such searches on their mobile devices during short gaps in time (e.g., between classes), and resume them later in a desktop environment when they can use the extra screen space and more convenient document storage capabilities of their computers. To support such searching, we have developed an academic digital library search interface that assists searchers in managing cross-session search tasks even when moving between mobile and desktop environments. Using a controlled laboratory study we compared our approach (Dilex) to a standard academic digital library search interface. We found increased user engagement in both the initial (mobile) and resumed (desktop) search activities, and that participants spent more time on the search results pages and had an increased degree of interaction with information and personalization features during the resumed tasks. These results provide evidence that the participants were able to make effective use of the visualization features in Dilex, which enabled them to readily resume their search tasks and stay engaged in the search activities. This work represents an example of how semi-automatic search task/session management and visualization features can support cross-session search, and how designing for both mobile and desktop use can support cross-device search.

CCS CONCEPTS

• **Information systems** → **Digital libraries and archives; Mobile information processing systems; Search interfaces; Human-centered computing** → **Empirical studies in HCI.**

KEYWORDS

Digital library search, mobile search interfaces, cross-session search, cross-device search, user study

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1 INTRODUCTION

With the seemingly continuous improvement of mobile device technology, there has been a shift away from using desktop environments for information access and toward using mobile devices [12]. While this pattern is also true for digital library search [7], there remains a preference for returning to desktop environments when the work goes beyond simple lookup tasks [13, 23]. Further, it is becoming increasingly common for searchers to use multiple devices across different search sessions to achieve their search goals [24], particularly when the search task is exploratory in nature [17]. We contend that with appropriate support built into the search interface, it is possible for digital library searchers to engage in initial search activities on their mobile devices at times that are convenient, and easily resume such tasks at a later date in their desktop environments.

To this end, we have designed and implemented an academic digital library search interface (Dilex) that supports cross-session search (XSS) and cross-device search (XDS). As re-finding previous search queries and saved documents are critical to enabling searchers to reacquaint themselves with previous progress [24, 30], we have designed Dilex to make such information readily available. We provide lightweight visual encoding of important contextual information, empowering searchers to identify, assess, and re-consider their prior search sessions.

Dilex uses a responsive design, sizing itself equally well for mobile and desktop environments. As the search activities are stored online rather than on the specific device used, any progress made on a search task on one device can be resumed on any other device with a simple ORCID iD login. As such, Dilex promotes a “search anywhere, anytime” ethos within the mobile setting, enabling work to be resumed and leveraged when returning to the search in a desktop environment. According to a recent taxonomy of the cross-device design space [5], this work falls within the following categories: temporally asynchronous, migratory between devices, single-user, personal scale, and mobile/semi-fixed dynamics.

Since Dilex was previously presented in a demonstration paper [14], we will only briefly outline its features in this paper. Our primary focus is the study of Dilex in comparison to a baseline academic digital library search interface, where we wish to answer

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the following high-level research question: What is the benefit of using Dilex for XSS and XDS within an academic digital library?

The remainder of this paper is organized as follows. Section 2 provides an overview of the literature that has informed this research. Section 3 outlines the key features of Dilex. Section 4 presents the specific research questions and methodology employed in the user study. Section 5 provides the results of the study organized by the research questions, along with a discussion of the implications and limitations. We complete the paper with a conclusion and outline of future work in Section 6.

2 LITERATURE REVIEW

While the type of academic digital library search being done can vary widely depending on the specific needs of the searcher, when the search spans multiple sessions it is commonly because of some element of complexity in the information need [25]. Compared to simple single-session fact-verification searching, undertaking complex multi-session searches often requires a higher-level of cognitive processing [25]. When such searches are also conducted cross-device, there is evidence that activities focused on learning (e.g., conducting broad searches) are important during the initial search, as they enable easy continuation when switching devices [44].

One model that excels at supporting higher-level cognitive processes and task preparation activities is the exploratory search model. Exploratory search generally consists of two phases: exploratory browsing, and then when sufficient certainty in the information need has been gained, focused searching [41]. When a searcher is in the exploratory browsing phase, they are generally concerned with discovering, learning, and investigating aspects of their information need, which are tasks that are aligned with the aforementioned preparation activities. When transitioning to focused searching, the activities change to (re)formulating queries, examining results, and extracting information.

As exploratory search processes are well-suited to supporting complex search tasks, and complex search tasks themselves are often undertaken over multiple search sessions [17, 24, 30], there is a synergy between exploratory search and XSS. In consideration of XDS, we assert that while the work of exploratory browsing may be done in either a mobile or desktop environment, the work of the focused search is more conveniently done in a desktop environment, where larger screens and physical keyboards enable multiple documents to be viewed simultaneously, which streamlines the information extraction process.

Consideration of why a searcher might stop a search session and resume it at a later date can help contextualize the need to support XSS. A recent study found that among the various task-based reasons for stopping (including having found all the needed information), 45% of the reasons provided were related to having to synthesize the information found and implied a specific need to continue the search later [24]. This same study found that among the non-task-based reasons, 74% of them were related to time limitations, fatigue, or being distracted. What we can conclude from this study is that the conditions that result in a XSS are dictated by the difficulty of the search task and external factors that keep the searcher from finishing it in one session.

When searchers resume a previous search session, it is important to know what they do to reacquaint themselves with their prior work, as this serves as a bridge from the previous search session to the current one. Studies have found that commonly used techniques are reviewing saved information, re-reading their own writing or notes, or re-issuing previous queries [24, 26, 30]. Unfortunately, the ability to accurately recall things like past queries is constrained [37]. Some studies have found that as many as 90% of search sessions include information resumption behaviour, wherein the searchers' activities are highly influenced by what they did in previous search sessions [43]. One particular study found that as many as 1/3 of searchers are unable to identify specific methods they use for reacquainting themselves with their prior search [24]. However, based on an evaluation of XSS tasks [16], searchers are likely to start the continuation of a previous search session by following a re-finding pattern of behaviour. Whether a searcher knows it or not, there is a lot of repetition of prior search effort when one resumes a search session, presenting an opportunity for the design of search interface to support search task resumption that minimizes the duplication of effort.

When searchers change to a different device between sessions, we characterize this as XDS. The factors involved in changing to a different device can be classified as geo-spatial, topical, or temporal [39]. Numerous studies have investigated the impact of geo-location on a searcher's device preference [9, 18, 38, 39, 45]. For example, a searcher's task may be interrupted as their work day ends, but they may use their mobile device to resume the search during their commute home [39]; or they may search a library index on their mobile device, but then use a computer when they arrive at the library [45]. The influence of the search topic on device preference is correlated with the complexity of the task. Mobile searchers tend to issue short queries and spend little time reading, often pursuing search tasks for which they have some prior knowledge [2]. They generally choose to search in broad domains, rather than conducting in-depth searches [9]. We contend that this is an issue of limited search support and the difficulty of saving information in mobile search environments. Temporally, there is a tendency for mobile search to be conducted early in the morning or late in the evening [29], when one might not want to put in the effort of accessing their computer.

A number of research efforts have been made to support XSS within the search interface. While digital workspaces are becoming increasingly common [11, 21, 40], generally they only allow a searcher to see what was previously found, but not how they got to these documents. The creation of tasks or sessions lowers the effort of resumption and re-finding by grouping relevant information together in the interface. Some have proposed approaches that automatically create task, query, or session groups [1, 30, 46], whereas others require the users to explicitly create and name tasks or groups of searches [6, 15, 27]. Context-based support for re-finding has been shown to be successful in systems where the user must provide explicit context [10], as well as when context is automatically detected [20]. Others have developed methods to support re-finding based on the degree of interaction with the saved documents [33]. The methods for presenting the session/task/query/document information to the searcher varies, including graph-based visualization

of the relationships between tasks or sessions [1, 46], visualization of the search/browsing timelines [20, 46], colour-encoding of saved documents [33], or hierarchical representation of queries and browsing histories [30].

With respect to cross-device support for search tasks, while there is a large body of research on the design space across many different application domains [5], there is very little research on this specific context. What has been done to support XDS has been limited to methods for sharing browsing history between devices [36] and re-ranking the search results on the desktop interface based on the mobile search history [17]. There is an opportunity for further study of whether some of the XSS approaches can support a responsive design that scales from mobile device to desktop environments.

3 DESIGN OF DILEX

We developed Dilex (Digital Library Explorer) [14] to facilitate cross-session academic digital library search tasks in both mobile and desktop contexts. Since the original demonstration of the approach, we refined the interface through heuristic evaluations [31] and cognitive walkthroughs [3]. As such, the interface shown in this paper is slightly different than the original version. Dilex provides semi-automatic search task/session management, and shows information hierarchically, providing increasingly specific information across three interfaces: the task dashboard, task detail, and SERP (see Figures 1 & 2, left, center, and right respectively).

3.1 Task Dashboard

From the task dashboard, searchers can create a new search task by issuing a query ❶, or resume previously suspended tasks. Each suspended task is represented by a task card ❷, which serves as a summary of the workspace, providing the task's current state and a timeline of search sessions attributed to that task ❸. When resuming a search task, clicking a task card opens the task detail interface and creates a new session. Clicking on the saved document count inside the task card opens the SERP interface and shows the documents saved for that task from all previous search sessions.

3.2 Task Detail

From the task detail interface, searchers can start a new search within the selected search task ❹, and browse previous sessions via the timeline ❺ or session cards ❻. Our goal with this interface was to provide insight into the progress of prior search sessions, thereby supporting task resumption (reacquaintance and re-finding) and scaffolding searchers as they refine and improve their queries.

Sessions are represented on the timeline by circles; selecting a circle highlights the corresponding session card with a dark border, and scrolling to a session card highlights the corresponding circle in the timeline. Selecting a query title opens the SERP interface and re-issues that query. Selecting the saved document count for the session displays the documents saved from that specific session.

3.3 SERP

The SERP interface allows searchers to browse the document cards ❸, and works in a similar manner regardless of whether the searcher has issued a new query or is viewing documents saved to the workspace. Searchers may select the documents they wish to save,

de-select documents they no longer wish to save, and gain insight into the topic they are searching via the displayed subject facets ❷. These design elements were specifically chosen to enable exploratory search, analysis, comparison, and learning.

As searchers scroll through the document cards, both the top-most document card and the facets associated with that document are highlighted with a dark border. Facets selected by the searcher are highlighted with a colour background ❹, the same colours are used to indicate which documents are associated with which facets ❺.

4 EVALUATION

In order to evaluate whether Dilex is effective at enabling XSS and XDS within an academic digital library context, we conducted a study comparing Dilex to a baseline search interface. The design of the study was motivated by the following seven research questions:

RQ1: What is the difference in user engagement during the initial (mobile) search when using Dilex versus the baseline?

RQ2: What is the difference in user engagement during the resumed (desktop) search when using Dilex versus the baseline?

RQ3: What kinds of search interface features are being used during the initial (mobile) search, when using Dilex versus the baseline?

RQ4: What kinds of search interface features are being used during the resumed (desktop) search, when using Dilex versus the baseline?

RQ5: What is the difference in the quality of the final set of search results after using Dilex versus the baseline?

RQ6: What is the difference in the time to task completion during each session, when using Dilex versus the baseline?

RQ7: What is the knowledge gain after each search session when using Dilex versus the baseline?

The baseline referenced in these research questions is a visually identical recreation of the search interface used by our University's digital library, but without the inclusion of some advanced query refinement and notification features that are beyond the scope of this study. Both interfaces use the same underlying search API, producing identical search results for a given query. Screenshots of this interface in both a mobile and desktop setting are shown in Figure 3.

4.1 Methodology

The research questions lead to a study design that has a single independent variable (digital library search interface) with two levels (Dilex, baseline). In order to avoid learning effects, we use a between-subjects design so that each participant is only exposed to one of the interfaces. There are four dependent variables in the study design: user engagement (RQ1 and RQ2), feature use (RQ3 and RQ4), quality of collected documents (RQ5), time to task completion (RQ6), and knowledge gain (RQ7).

The user engagement was measured after each search session, using the User Engagement Scale - Short Form (UES-SF) instrument [32]. UES-SF is composed of four dimensions: focused attention (FA), perceived usability (PU), aesthetic appeal (AE), and reward factor (RW). For each dimension, a set of three statements are provided and participants are asked to rate their level of agreement on

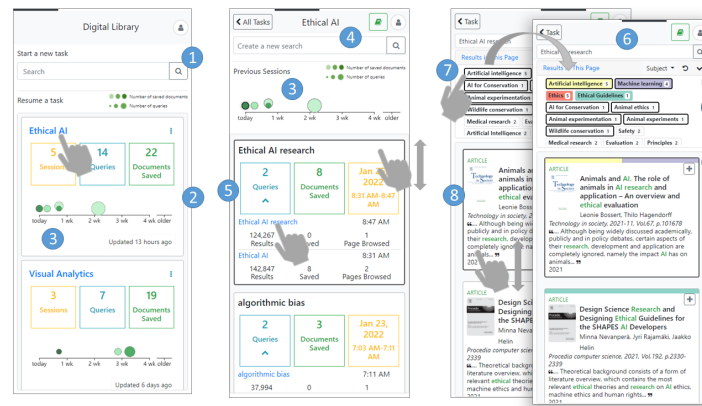


Figure 1: Dilex interface elements (mobile view): Task dashboard (left), task detail (center), SERP facet unselected (right background), SERP with three facets selected (right foreground)

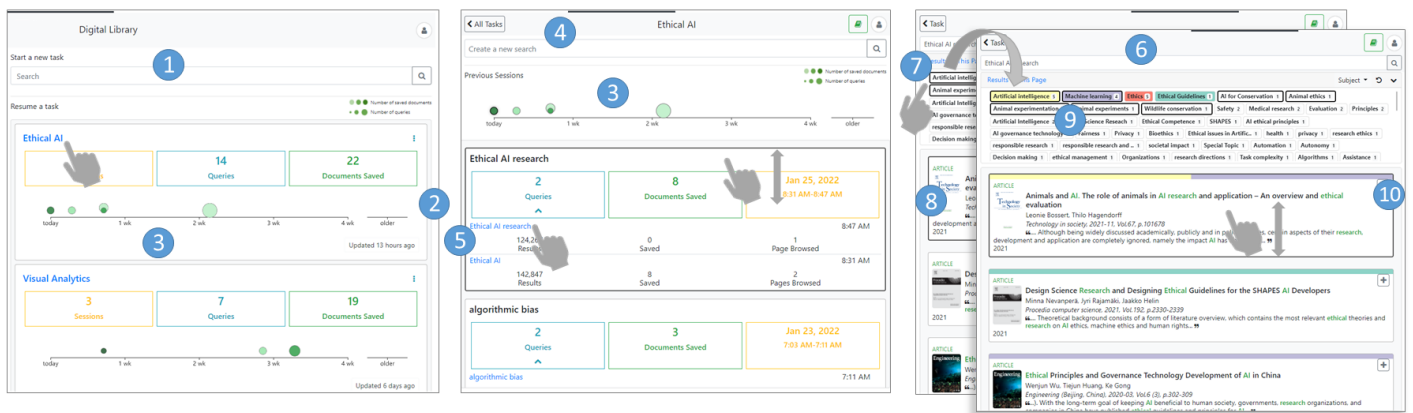
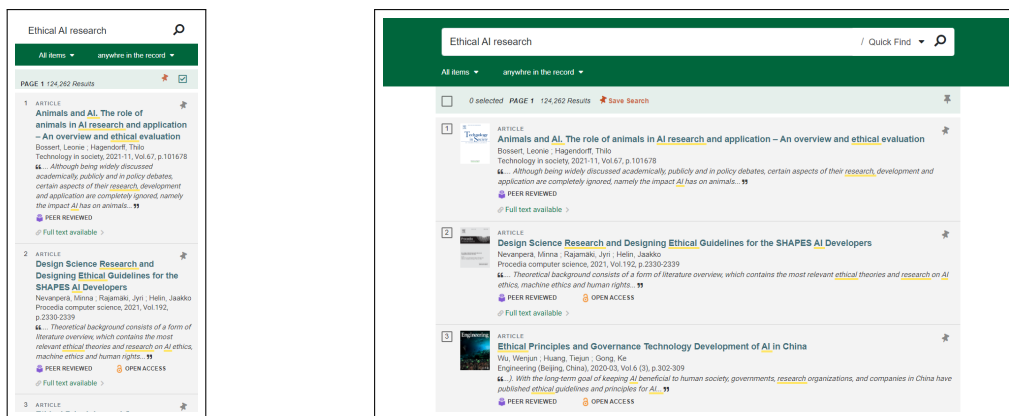


Figure 2: Dilex interface elements (desktop view): Task dashboard (left), task detail (center), SERP facet unselected (right background), SERP with three facets selected (right foreground)



(a) Baseline mobile view

(b) Baseline desktop view

Figure 3: Screenshots of the mobile and desktop view of the baseline interface.

a five-point Likert scale. We use the mean values of the responses for each participant with respect to each of the four dimensions, as well as the overall engagement (e.g., all 12 measures).

In order to measure the feature use, we logged all interaction on both user interfaces using LogUI [28]. We started by classifying and counting the use of each interactive element according to Wilson's taxonomy of search interface features [42]: input, information, control, and personalization. The input features included any that allowed a searcher to provide specific input to the search process (issuing a query). The information features included any that provided a searcher with information about a search result (viewing the SERP, clicking through to a document). The control features included any that enabled a searcher to manipulate the views (pagination controls). The personalization features included any that allowed a searcher to customize the interface to their specific needs (saving a document or query, document tagging/subject facet highlighting). Since the personalization features also included the ability for participants to use input, information, and control features from the personalized views, we created three additional interface feature classes and assigned the interactive features from the personalized views to these: personalization-input (re-issuing a query from the workspace), personalization-information (re-visiting a document from the workspace), and personalization-control (pagination or navigation within the workspace). Other than the document tagging (baseline) and subject facet highlighting (Dilex), both interfaces provided a similar set of interaction mechanisms across all interface feature types.

The quality of the collected documents was measured by extracting the set of documents saved to the workspace at the end of the resumed search task, performing relevance assessments with respect to the assigned task, and calculating per-participant precision. The relevance assessment was conducted using a three-point scale (2=fully relevant; 1=partially relevant; 0=not relevant). Two of the authors independently assessed twenty of the saved search results, met to discuss and re-calibrate their interpretation of the relevance scales, and then assessed another twenty of the documents. They then re-assessed the first twenty as well as all that remained. The inter-rater reliability of their relevance scores was calculated using Krippendorff's alpha [22]. With $\alpha = 0.73$, we can classify this as substantial agreement, noting that there was disagreement about the relevance of some documents (most of which were in relation to whether the partially relevant score should be applied or not). To address this issue, the third author carefully assessed each document for which there was disagreement, serving as the tie-breaker. These relevance assessments were collapsed into binary scores, with the partially relevant documents considered not relevant [35]. Doing so facilitated the calculation of the precision scores (number of relevant documents divided by the total number of documents in the workspace) for each participant.

The measure of time to task completion was extracted from the interaction logs. The time metrics were calculated at two different levels of granularity: (1) the time to task completion for the entire initial and resumed search sessions; and (2) the time spent in each of the two states of the search interfaces: query/SERP and personalization/workspace.

Knowledge gain was measured using a vocabulary test administered at multiple times throughout the search process: before the

initial search, after the initial search, before the resumed search, after the resumed search. Following a process similar to that of Roy et al. [34], we provide ten phrases about the primary search topic, and asked the participants to provide definitions for each phrase. Five of these phrases were discovered to be poorly supported by the available search results, and were dropped, leaving five for analysis. We prepared a grading rubric for each definition (-1 for no answer; 0 for an answer that is incorrect, 1-3 for partially correct answers, and 4 for a fully correct answer), and followed a similar procedure to what was described above for the relevance assessments. The inter-rater reliability (Krippendorff's alpha) for each vocabulary phrase was: "AI Ethical Guidelines" $\alpha = 0.43$ (moderate agreement); "AI Ethical Issues" $\alpha = 0.54$ (moderate agreement); "AI4People" $\alpha = 0.62$ (substantial agreement); "Bias in AI" $\alpha = 0.46$ (moderate agreement); and "Meaningful Human Control" $\alpha = 0.65$ (substantial agreement). The overall inter-coder reliability was $\alpha = 0.58$, which is considered moderate agreement. All cases of inconsistent coding were carefully assessed by the third author, who served as the tie-breaker.

4.2 Study Procedures

The study itself was structured to span two separate sessions: the initial session conducted on a mobile device, and the resumed session conducted in a desktop environment on a computer. A gap of approximately one week between the search sessions (seven to nine days, depending on the participants' schedules) was used to ensure that the task reflected a realistic cross-session scenario. This gap is consistent with other studies on cross-session search [8, 30]. The study sessions were all conducted remotely over Zoom, with the participants using their own mobile devices and desktop/laptop computers. See Figure 4 for an illustration of the overall process.

The initial search session started with the process of obtaining informed consent to participate in the study. Demographic information about the participants was collected using an online questionnaire, including their search experience with digital libraries, their use of different device types for such searching, and their cross-session search behaviour. Participants were provided with a

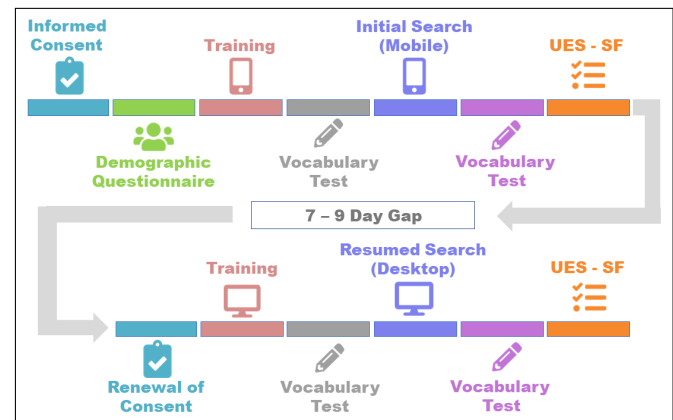


Figure 4: Structure of the user study.

training video that focused on mobile device use for the assigned search interface (Dilex or baseline), and were given an opportunity to perform this task on their own mobile device in order to familiarize themselves with the interface features. Next, the simulated search task (which will be explained in the following sub-section) was introduced and the vocabulary test was administered. Participants were asked to proceed with the initial search task, with a 10 minute limit imposed to simulate an external constraint and the need to resume the search at a later date. At the completion of the task, the vocabulary test was administered again, followed by the user engagement (short form) questionnaire. The follow-up session was scheduled and participants were asked not to search about the topic in the intervening time.

The resumed search session started with the renewal of consent to continue with the study. The remainder of the process mimicked the initial search session, including the use of a training video that focused on desktop environment use of the assigned search interface, the opportunity for the participants to use the interface on their own desktop or laptop computer, the explanation of the resumption of the simulated search task, the administering of the vocabulary test, the actual resumption of the search task (for which there was no explicit time limit), the administering of the final vocabulary test, and administering the final user engagement questionnaire. After all data collection was complete, participants were compensated for their time via a \$50 e-transfer.

4.3 Simulated Search Task

As this study focused on XSS, the simulated search task was prepared in two parts: the initial search and the resumed search. For the XDS aspect, we considered two alternatives: (1) start the search on a mobile device and resume it in a desktop environment, or (2) start the search in a desktop environment and resume it on a mobile device. Since prior research has identified the first of these patterns to be common in library search [45], we limited our study to this device user order.

More specifically, we envisioned a scenario where a searcher would start their search task on their mobile device because they have limited time and their computer is not readily available, and later resume this task when they can use their computer in a desktop environment. Further, we considered that this scenario would match well with the exploratory search model [41], with the initial search following an exploratory browsing pattern (learning about the domain) and the resumed search following a focused searching pattern (issuing focused queries and assessing relevant documents). This is consistent with prior research that found mobile searchers have a tendency to avoid tasks that they perceive to be difficult [2], and therefore may save the difficult aspects of such a task for when they return to a desktop environment. This scenario formed the basis for the design of the search task, following Borland's simulated work task framework [4]. See Figure 5 for the details and instructions given to the participants regarding the search task used in this study.

4.4 Participants

Thirty (30) participants were recruited for this study, with half assigned to each of the two experimental conditions (using either

Search Context: Suppose you are taking a course on Computer Ethics and recently you have received an assignment to write a paper on the topic of Ethical AI. The rise of AI in recent years has transformed our lives. The use of AI in sectors such as business, education, healthcare, and security has automated many human processes. However, without proper monitoring, malicious uses of AI technology can threaten data security and online privacy. For the assignment, you need to write a paper on the current state of Ethical AI research, citing relevant literature on this topic.

Initial Search Instructions: Imagine you are at the university and have a 10-minute break between two classes. Since you are busy, you want to use this time effectively, and have decided to start searching for a few academic articles on this topic to get started.

Since you are away from your desktop computer, you will use your smartphone to do this search task. Your goal is to learn about the topic by conducting a series of searches, reviewing the search results, and saving any documents you would like to consult later when you get back to the task on your computer. You have a time limit of 10 minutes for this task.

Resumed Search Instructions: About a week ago, in a gap in time between classes, you spent about 10 minutes searching for academic articles to support the paper you are going to write about Ethical AI. The use of AI in sectors such as business, education, healthcare, and security has automated many human processes. However, without proper monitoring, malicious uses of AI technology can threaten data security and online privacy.

Since you are at your home and you have your computer with you. You now want to resume your previously unfinished search task of finding relevant academic documents for your paper. Since your prior search activities focused on learning about the topic, your goal now is to pick up where you left off, and then focus on a specific aspect of Ethical AI so that your paper can have a narrow focus. Continue searching for academic articles, save what you like, and delete those that are no longer relevant.

Figure 5: The assigned cross-session cross-device search task provided to the participants.

Dilex or the baseline search interface). In order to avoid a potential confounding factor with the level of experience with academic digital library search interfaces, we focused our participant recruitment on graduate students who would have an intermediate to high level of experience with such academic searching. Since the topic of the search task is related to computer science, we further focused our participant recruitment on graduate students in this field of study, which created a limitation for participant recruitment.

The participants represented a relatively coherent population of academic digital library searchers. The median frequency of academic searching on a mobile device was in the 1-3 times per month range, whereas the median frequency of academic searching in a desktop environment was in the 4-6 searches per week range. When asked about their XSS/XDS behaviour, the participants reported that when they started their session on a mobile device, they were more likely to resume it in a desktop environment, which matched the scenario used in our study design.

4.5 Analysis

As the research questions and resulting study design focused on the comparison of Dilex to the baseline search interface, we have organized the analysis of the results around these research questions and this specific comparison. All data collected in this study was numerical in nature: mean user engagement scores for RQ1 and RQ2, counts of feature use by category for RQ3 and RQ4, precision of the saved documents for RQ5, time to task completion for RQ6, knowledge gain for RQ7. The measurements of knowledge required special processing to represent knowledge gain. For this purpose, we calculated the difference between the measures made at specific stages of the study.

For the statistical analysis of the differences between the interfaces, ANOVA tests were conducted using a p-value of 0.05 to denote significance. Given that counts of feature use (RQ3 and RQ4) can be skewed, the non-parametric Mann-Whitney-Wilcoxon test was employed to analyze this data.

One participant in the baseline condition exhibited behaviour that was indicative of misunderstanding the assigned task instructions and study procedures. The number of documents they had saved at the end of the resumed task was more than three standard deviations from the mean value. It appeared that they were collecting as many documents as they could, with a plan of re-considering their relevance at a later date. Given that this behaviour affected the other measures in this study, we removed all data from this participant in the analysis, resulting in a comparison of 15 participants who used Dilex to 14 participants who used the baseline.

There was one further complication in the data analysis. The software used to log the timestamp when participants transitioned back and forth between the query/SERP and personalization views in Dilex was misconfigured for the first four participants in the resumed (desktop) portion of the study. As a result, this aspect of the time to completion analysis could only be conducted for 11 of the Dilex participants compared to the 14 baseline participants. We acknowledge that these issues have resulted in the study being slightly under-powered.

5 RESULTS AND DISCUSSION

5.1 RQ1: User Engagement During Initial Session

Comparing the user engagement scores between the participants who used the two different academic digital library search interfaces (Dilex, baseline) during their initial search on a mobile device revealed statistically significant differences in the overall measure as well as in all four dimensions (see Table 1). In all cases, the user engagement scores were higher for those participants who used Dilex. This result shows that mobile version of the Dilex interface enabled the searchers to be more engaged in the initial search task than the baseline mobile interface. We attribute this to both the high quality mobile-friendly design of Dilex (perceived usability, aesthetic appeal), as well as the ease with which searchers can keep track of their task, session, query history, and saved documents during the exploratory browsing process (focused attention, reward factor).

5.2 RQ2: User Engagement During Resumed Session

The comparison of the user engagement scores between the participants who used the two different search interfaces during the resumed search in their desktop environments showed statistically significant differences in the overall user engagement as well as in all four dimensions of the measure (see Table 1), all in favour of Dilex. We attribute the superior user engagement with Dilex to the visually engaging design (perceived usability, aesthetic appeal) as well as the support built into the system that allows searchers to easily see what past searches were done (including the queries issued) and what documents were saved from each query, enabling them to readily pick up where they had left off with the initial search (focused attention, reward factor).

5.3 RQ3: Feature Use During Initial Session

Comparing how frequently each of the different types of interface features were used during the initial (mobile) search session revealed some noteworthy similarities and differences between the participants, depending on whether they were assigned to Dilex or the baseline. Participants who used the baseline issued slightly more queries (input features) than those who used Dilex and went slightly deeper into the search results (control features), but these differences were not found to be statistically significant. Participants who used Dilex accessed the details for more of the search results, which was reflected in their use of information features (viewing search result details), which was statistically significant.

The personalization features included a combination of saving or removing documents, as well as manipulating those documents using document tagging (baseline) or subject facet highlighting (Dilex). These features were used intensively in both interfaces, with slightly more interactions in Dilex, but not at a statistically significant level. While issuing queries from within the personalized elements of the interfaces (personalization-input features) or accessing the saved documents (personalization-information features) was used lightly, this is expected due to the time-limited exploratory browsing nature of the task. Both the baseline and Dilex participants did some navigation within the personalized views (personalization-control features), slightly more so with Dilex but not at a statistically significant level.

5.4 RQ4: Feature Use During Resumed Session

The degree of feature use between the two interfaces during the resumed (desktop) session followed a somewhat similar pattern to that of the initial (mobile) search. Participants issued a similar number of queries, and there was a minimal difference in how deeply they delved into the search results (input and control features, respectively; not statistically significant). However, the Dilex participants accessed the details of documents using the information features nearly four times as much (statistically significant).

Further, there were noteworthy differences in how the participants used the personalization features. In the resumed search session, the Dilex participants interacted with the core personalization features (saving documents to the workspace) on average six more times than the baseline participants, but this difference was not at a statistically significant level. The Dilex participants re-issued

Session	UES-SF Dimensions	Baseline	Dilex	ANOVA
Initial Session	Focused Attention (FA)	3.83 (0.581)	4.31 (0.541)	F(1,27) = 5.255, p <0.05
	Perceived Usability (PU)	3.60 (1.13)	4.42 (0.462)	F(1,27) = 6.788, p <0.05
	Aesthetic Appeal (AE)	3.14 (0.713)	4.42 (0.648)	F(1,27) = 25.63, p <1 x 10⁻⁴
	Reward Factor (RW)	4.02 (0.497)	4.67 (0.454)	F(1,27) = 13.24, p <0.005
	Overall User Engagement	3.65 (0.486)	4.46 (0.429)	F(1,27) = 22.51, p <0.0001
Resumed Session	Focused Attention (FA)	3.88 (0.883)	4.60 (0.507)	F(1,27) = 7.36, p <0.05
	Perceived Usability (PU)	4.05 (0.783)	4.67 (0.356)	F(1,27) = 7.688, p <0.01
	Aesthetic Appeal (AE)	3.43 (0.789)	4.27 (0.737)	F(1,27) = 8.75, p <0.01
	Reward Factor (RW)	4.26 (0.573)	4.87 (0.246)	F(1,27) = 14.00, p <0.001
	Overall User Engagement	3.90 (0.590)	4.60 (0.358)	F(1,27) = 14.95, p <0.001

Table 1: Mean values (standard deviation) for the user engagement scores collected after using the assigned search interface in the initial search session and the resumed search session (5-point Likert scale; larger values represent greater engagement score).

Session	Feature Type	Baseline	Dilex	Mann-Whitney-Wilcoxon
Initial Session	Input	4.86 (4.26)	3.47 (2.33)	W = 127.5, p = 0.330
	Information	9.93 (11.6)	21.2 (13.8)	W = 53, p <0.05
	Control	4.36 (5.93)	2.73 (2.49)	W = 107.5, p = 0.930
	Personalization	13.4 (9.50)	15.9 (8.72)	W = 75.0, p = 0.196
	Personalization-Input	0.07 (0.27)	0.47 (1.06)	W = 84.0, p = 0.173
	Personalization-Information	0.14 (0.54)	0.53 (1.19)	W = 91.0, p = 0.325
	Personalization-Control	2.14 (3.37)	3.53 (7.57)	W = 96.5, p = 0.714
Resumed Session	Input	3.07 (3.22)	3.09 (2.02)	W = 64.0, p = 0.484
	Information	6.21 (7.18)	23.7 (9.53)	W = 13, p <0.0005
	Control	1.93 (1.94)	1.64 (1.63)	W = 78.5, p = 0.955
	Personalization	9.79 (5.73)	16.1 (12.5)	W = 46.5, p = 0.100
	Personalization-Input	0.57 (0.94)	1.45 (0.93)	W = 31, p <0.01
	Personalization-Information	7.07 (5.68)	3.00 (3.46)	W = 110.0, p = 0.07
	Personalization-Control	4.71 (3.75)	7.64 (3.83)	W = 40.0, p <0.05

Table 2: Mean values (standard deviation) of the number of user interactions for each feature type during both the initial and resumed search sessions for all feature usage types during both search sessions (larger values represent greater interaction).

their queries from the personalized views (personalization-input) and navigated among these views (personalization-control) more frequently than the baseline participants (statistically significant). Conversely, the baseline participants re-accessed previously saved documents (personalization-information) more often than the Dilex participants (not statistically significant).

These exemplify the different approaches participants took while reacquainting themselves with their previous search activities. The Dilex participants navigated and explored among the previously saved information. They were able to easily find and access their prior queries and even consider which ones to re-issue since they could see what documents were found with each. The baseline participants made minimal use of the search history and instead preferred to re-visit saved documents to remind themselves of what they had found.

5.5 RQ5: Quality of Collected Documents

While the total number of documents saved to the workspace at the end of the study was greater for the participants using Dilex than those who used the baseline, the difference was not found to be statistically significant (see Table 3). Among these documents, the precision of those saved by the participants using the baseline

was slightly higher than those saved by the Dilex participants; the difference was not statistically significant either.

Given that the participants were given the same exploratory search instructions and were exposed to the same set of search results, it is not unexpected that the quality of the collected documents would not be affected by the interface used; both groups of participants were successful in finding a reasonably large collection of relevant documents for the assigned task. It is encouraging to see that the Dilex participants saved on average two more documents, which is consistent with their increased use of the information features in the interface (RQ3 and RQ4).

5.6 RQ6: Time to Task Completion

The time to task completion for the initial session was nearly identical for all participants in both experimental conditions (Dilex v. baseline), due to the 10 minute time limit imposed on that session to create a realistic need to resume the search at a later date. There were only minimal differences in how much time the participants spent with the primary query/SERP interface and the personalization views (see Table 4). The small amount of time spent in the personalization views was consistent with the feature use during the initial session (RQ3) and the expectations of a time-limited exploratory browsing task where the searchers will primarily be

Session	Measure	Baseline	Dilex	ANOVA
Resumed Session	Saved Documents	9.2 (5.8)	11.9 (5.6)	F (1, 27) = 1.64, p = 0.207
	Precision	0.85 (0.13)	0.78 (0.24)	F (1, 27) = 0.789, p = 0.382

Table 3: Number and precision of the saved documents (standard deviation) for both interfaces at the end of the resumed session.

Session	State	Baseline	Dilex	ANOVA
Initial Session	Query/SERP Time	9:20 (1:17)	9:22 (1:42)	F(1, 27) = 0.002, p = 0.968
	Personalization/Workspace Time	0:53 (1:02)	0:56 (1:43)	F(1, 27) = 0.006, p = 0.939
Resumed Session	Time to Task Completion	13:39 (6:46)	18:03 (5:28)	F (1, 27) = 3.729, p = 0.064
	Query/SERP Time	7:27 (7:19)	13:34 (6:04)	F(1, 23) = 4.989, p < 0.05
	Personalization/Workspace Time	6:13 (4:21)	5:23 (3:35)	F (1, 23) = 0.259, p = 0.616

Table 4: Mean values (standard deviation) of the time (minutes:seconds) spent in the query/SERP and personalization/workspace views for both interfaces and both sessions.

Comparison	Before (All Participants)	After (All Participants)	ANOVA	Knowledge Gain (Baseline)	Knowledge Gain (Dilex)	ANOVA
before/after initial search	-0.29 (0.51)	-0.11 (0.54)	F(1, 28) = 8.20, p < 0.01	0.14 (0.42)	0.23 (0.28)	F(1, 27) = 0.41, p = 0.53
before/after resumed search	0.01 (0.62)	0.53 (0.68)	F(1, 28) = 27.6, p < 1 x 10⁻⁴	0.36 (0.53)	0.67 (0.50)	F(1, 27) = 2.61, p = 0.12

Table 5: Mean knowledge scores (standard deviation) compared before and after the initial and resumed searches (range: [-1, 4]), and knowledge gain over the session grouped by the interface used.

searching and saving documents that appear to be relevant, but not necessarily spending much time reviewing those saved documents.

For the resumed session, there was no time limit on the task. While the Dilex participants spent more time overall, the comparison to those who used the baseline was not statistically significant. If we compare the time spent viewing the primary query/SERP interface and the time spent in the personalized views, we find interesting differences. The Dilex participants spent more time within the query/SERP mode. Since the number of queries issued were not different between the Dilex and baseline participants (see the analysis of RQ4), we conclude that the extra time in the query/SERP mode was spent evaluating and viewing search results, which we can see with the significantly higher number of information features being used by the Dilex participants (also identified in RQ4) and the larger number of saved documents (see RQ5). With respect to the personalization views, the Dilex and baseline participants spent a more or less equivalent amount of time here, showing that they were equally efficient at the task resumption activities. This is interesting, since they took decidedly different interactive approaches to this (Dilex participants re-issuing queries via the personalization-input features in Table 2 v. baseline participants re-visiting search results via the personalization-information features).

5.7 RQ7: Knowledge Gain

Our primary interest regarding knowledge gain is to determine whether the initial and resumed tasks resulted in changes in knowledge levels, and whether there was an effect due to the interface used. These results are reported in Table 5. Here we can see that while overall the knowledge levels are low (-1 = no answer; 1-3 =

partially correct answer), there are statistically significant differences after both the initial and resumed search, when we consider all of the participants together. When we calculate the knowledge gain and compare the participants who used the baseline to those who used Dilex, we see greater knowledge gain from the Dilex participants, but not at a statistically significant level.

Like the quality of the search results (RQ5), since the participants were exposed to the same sets of search results, it is not unusual to find minimal difference in knowledge gain as a result of the interface used. Both interfaces provided similar support for saving search results and later returning to re-assess the saved documents. What is important is that the participants as a group increased their knowledge as a result of the initial and resumed searches, which is a sign that they took the search activities seriously, contributing to our confidence in the other findings from this study. They also issued many newly formulated queries which also took them to the same path. Later, they spent a significant amount of time looking at document information and document sources to reacquire themselves with their past search activities and discover new information according to address their information need.

5.8 Discussion

Overall, these findings paint a picture of how Dilex supports users in XSS in the context of a specific XDS scenario: starting from a mobile device and resuming in a desktop environment. During the initial (time-limited) search on a mobile device, participants made significantly more use of the information elements in Dilex, illustrating their willingness to engage with the search process using this interface. This was reflected in the significantly higher scores on all dimensions of user engagement. The extra information

provided in the mobile version of Dilex enabled the searchers to clearly see, track, and monitor the query history in the session and the documents found during this exploratory browsing activity.

After returning to the search task and changing devices to continue the task in their desktop environment, there were further differences between the baseline and Dilex participants that illustrate the XDS benefits of Dilex. While both groups of participants were equally efficient in using the personalization features to reacquaint themselves with their prior search task, the baseline participants primarily did this by re-visiting previously saved documents, whereas the Dilex participants navigated among what was saved and re-issued past queries. We attribute this difference to the Dilex features that enable grouping and visualizing the outcomes of past queries and linking these to the documents saved from each, which could easily be observed and used with the larger screen sizes in the desktop environment. When the search task was continued, the Dilex participants spent significantly more time in the query/SERP mode. As with the initial search, the Dilex participants continued to use more of the information features, and reported significantly higher scores in user engagement. While these additional interface features ultimately had minimal effect on increasing the participants' abilities to find relevant documents or learn about the topic, the high user engagement scores, especially in relation to the reward factor dimension, highlight the perceived benefit of using Dilex.

The search task used in this study was complex in nature, requiring a high-level of cognitive processing similar to what has been found in other studies of XSS [25]. We asked the participants to conduct initial searches on the mobile device to encourage exploratory browsing and support learning, which is consistent with what some have found happens already [44]. We placed a time limit on the initial search, which is a common non-task-based reason reported for stopping a search task [24]. When returning to the search task, the features of both interfaces provided access to critical reacquaintance information such as previous queries and saved information [24]. Given that the study design is consistent with what searchers do in practice, we are confident in the ecological validity of the findings from this controlled laboratory study.

While this study was conducted in the context of academic digital library search, the results can be generalized to other search contexts where a searcher cannot reasonably be expected to complete their search task in a single session, with some caveats. We expect similar results in situations where it is convenient to initiate exploratory browsing on a mobile device (such as when commuting [39]), and resume the task later in a desktop environment to conduct focused searching. Whether the benefits of using Dilex hold for the reverse XDS order, or when searching in collections that have more heterogeneous data than academic digital libraries, remains unclear.

The primary limitation of this work is in relation to the controlled aspect of the study. We specified a particular information need and defined a specific scenario of cross-session cross-device searching. While this level of control allowed us to make direct comparisons between the two groups of participants in this study, it limits the external validity. Further, since the study was conducted remotely due to COVID-19 protocols, using their own mobile devices and desktop computers may have introduced potential confounding factors

such as screen size and computing power. These same differences, however, improve the ecological validity of the study.

6 CONCLUSION AND FUTURE WORK

The primary contributions of this research are the design of the Dilex interface which uses visualization techniques to support the monitoring of the task progress over multiple search sessions, the implementation of this approach to support both mobile and desktop search, and the results of the study that illustrate the benefits of this approach. This work represents an example of how lightweight information visualization techniques can be used to enable interactive information retrieval processes [19].

While the results of this study have given us confidence in the ability for Dilex to support XSS and XDS within an academic digital library, further study in a naturalistic setting is needed. For future work, we plan to deploy Dilex broadly within our University and conduct both log analyses and qualitative studies to assess how searchers are able to use this interface within their own academic digital library search practices.

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